From a loss law to a model-size decision

This companion expands notes §§1.1–1.3 and 8.1, and blog §8.1. The central distinction is between fitting a loss surface and choosing an allocation under constraints. All calculations below are derivations or explicitly illustrative examples, not newly measured results.

Derive the allocation, then check its assumptions

For $L = E + AN^{-\alpha} + BD^{-\beta}$, assume $A, B, \alpha, \beta > 0$ and approximate training compute by $C = 6ND$. Here N is a consistently defined parameter count, D is training tokens, and L is common-distribution validation loss. Substitute $D = C/(6N)$:

$$L(N; C) = E + AN^{-\alpha} + B(6N/C)^{\beta}.$$

Differentiating with respect to $\log N$ gives

$$-\alpha AN^{-\alpha} + \beta B(6N/C)^{\beta} = 0.$$

Thus the optimum balances the weighted model and data contributions:

$$N_* = \left(\frac{\alpha A}{\beta B}\right)^{1/(\alpha+\beta)} (C/6)^{\beta/(\alpha+\beta)}, \qquad D_* = C/(6N_*).$$

The model and token contributions need not be equal unless their exponents are equal. The floor E drops out of this allocation calculation but still affects predicted loss and equivalent-compute ratios. This algebra assumes a continuous size axis, no memory/serving constraints and the stated FLOP approximation. Actual architectures form a discrete feasible set. Hoffmann et al., Chinchilla, §3.

What an outer coupling exponent changes

Write $f = AN^{-\alpha} + BD^{-\beta}$ and $L = E + f^k$ with $k > 0$. Minimizing L is equivalent to minimizing f **holding its fitted coefficients fixed**. That preserves the algebraic allocation formula. A separate fit can change A, B, α, β and therefore the numerical optimum.

For $0 < k < 1$,

$$\frac{\partial^2 L}{\partial N \partial D} = k(k-1) f^{k-2} \alpha\beta AB N^{-\alpha-1} D^{-\beta-1} < 0.$$

This expresses complementarity: increasing data strengthens the marginal effect of increasing size in this functional form. It does not establish that all observed recipes have this interaction. Compare competing forms on development slices and inspect residual structure. Videau et al., Skaling, model formulation.

Translate a loss difference into compute

For a baseline curve $L(C) = E + AC^{-\gamma}$ and a positive improvement $\delta < L - E$, reaching $L - \delta$ on the baseline requires

$$M = \frac{C_{\text{equiv}}}{C} = \left(\frac{L - E}{L - E - \delta}\right)^{1/\gamma}.$$

An illustrative $\gamma = 0.05$, $L - E = 1$ and $\delta = 0.01$ give $M \approx 1.22$. The same raw loss difference can therefore imply a large compute gap. This is a model-dependent estimate: uncertainty in E or γ must accompany it. A multiplier is not automatically the same fraction of wall-clock savings.

Make an allocation decision with uncertainty

For each bootstrap fit, solve over the actual feasible architecture/token grid. Retain the distribution of chosen sizes, candidate gaps and loss regret. If many nearby sizes are statistically indistinguishable, serving cost or training throughput can decide among them; the point estimate alone should not dictate a flagship size.

Required artifact: a surface fit, a constraint-aware optimum, neighboring feasible configurations, and uncertainty on the decision. Continue with fitting and experimental design.

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